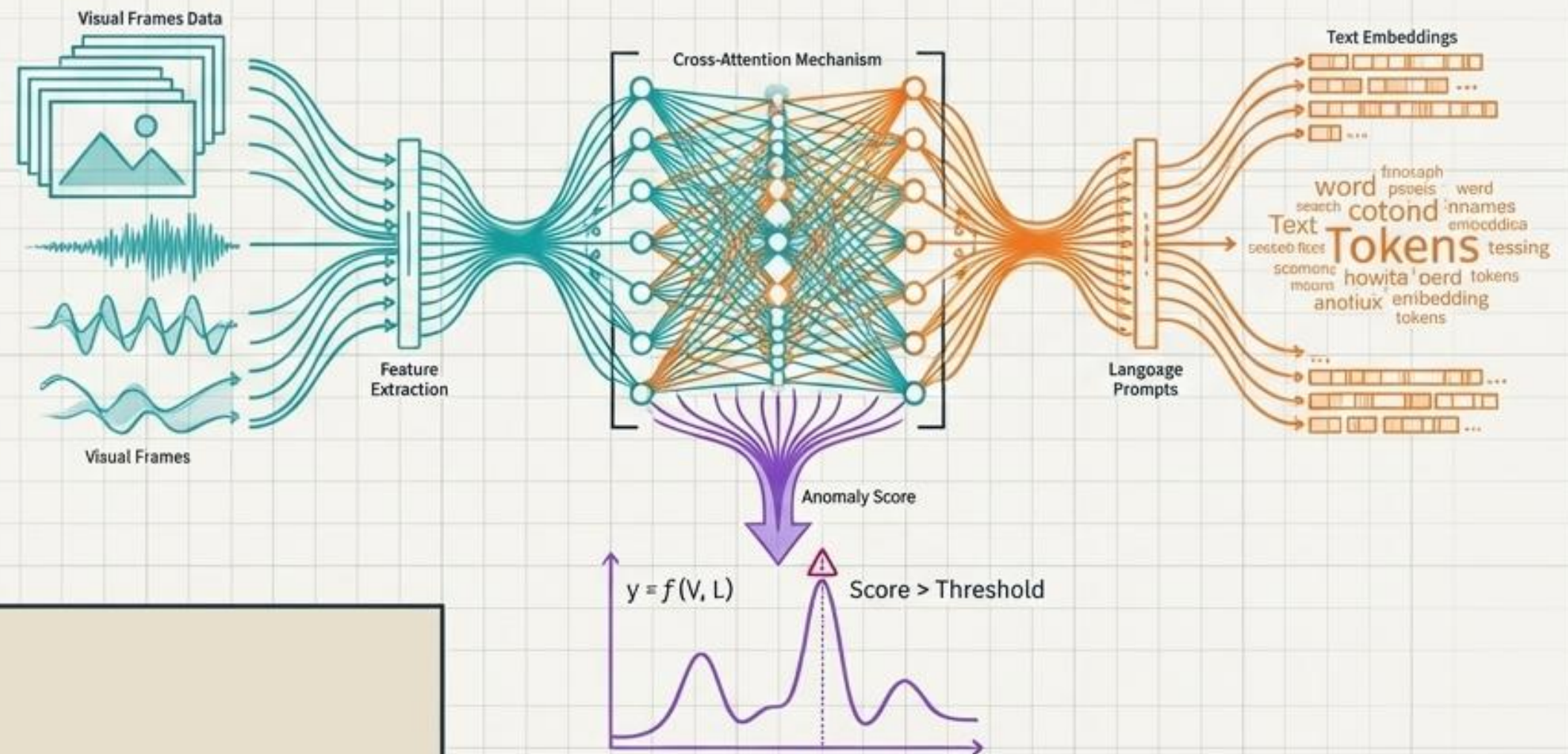


Weakly Supervised Video Anomaly Detection via Language-Guided Cross-Attention

Applied Machine Learning Progress Report: V1 Milestone & V2 Architecture Roadmap



Presenters: Group 18 |

Date: 20 March 2026

The V1 Milestone: Surpassing Foundational Baselines with a Semantic Lens

Panel 1: The Breakthrough (V1 Performance)



Video-Level AUROC: 94.85%
(Exceptional anomaly identification)

Frame-Level AUROC: 77.14%
(Precise temporal localization)

2. The Benchmark

Baseline (Sultani CVPR 2018)

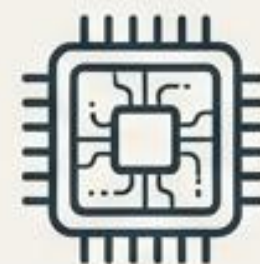
75.41%

Our Status

77.14%

Successfully surpassed the foundational Multiple Instance Learning (MIL) benchmark using a novel modality fusion approach.

Panel 3: The Efficiency



Model Size: 2.17M Parameters
(vs. 30M+ for C3D baselines)

Computational Load:
138.68M FLOPs per inference

Training Time:
~20 minutes / 100 epochs
(RTX 4060)

Overcoming Context Bias in Traditional Visual-Only Models



NORMAL



ANOMALY



≠



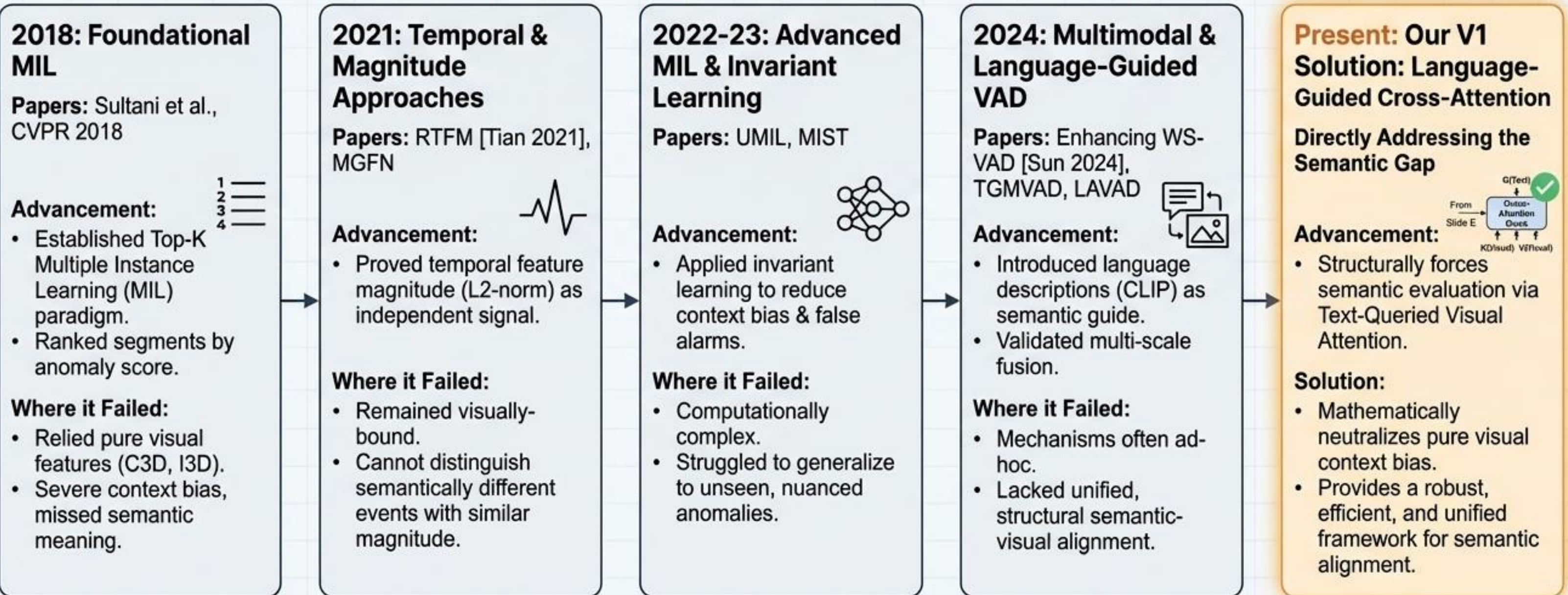
The Failure Mode of Traditional MIL

Current state-of-the-art models rely entirely on visual extraction (C3D, I3D). They suffer from **Context Bias**: evaluating pixel variation rather than semantic meaning (e.g., a running commuter vs. a fleeing suspect).

Language as a Semantic Filter

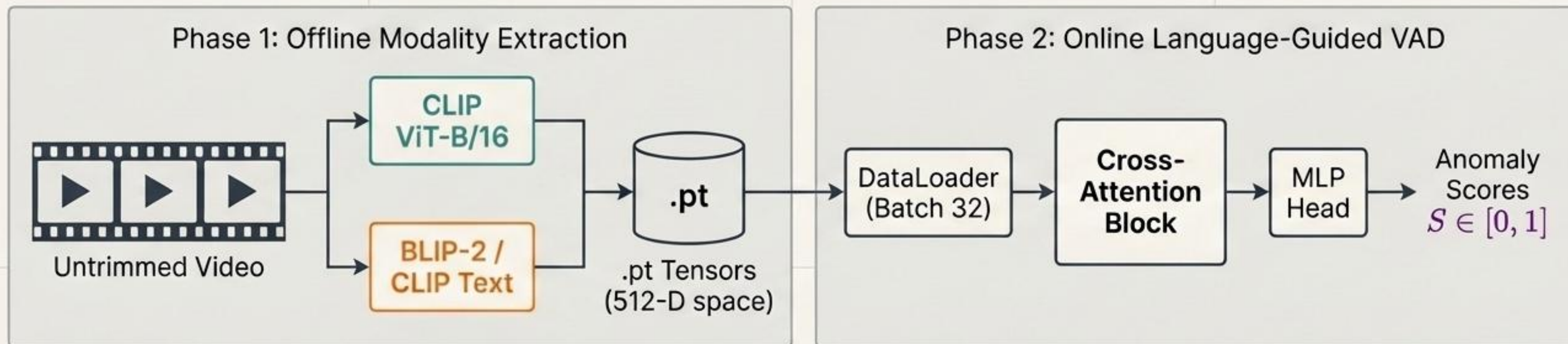
By introducing natural language descriptions of video segments as a semantic guide, we force the model to evaluate the meaning of an event, independent of its visual appearance.

The Chronological Evolution of WS-VAD: Mapping Advancements, Failures, and the Path to V1



Synthesis Takeaway: The V1 model logically evolves from the field's failures, providing the missing structural mechanism for true semantic anomaly detection.

Phase 1 & 2: The Offline-Online Modality Fusion Pipeline



Phase 1: Modality Extraction

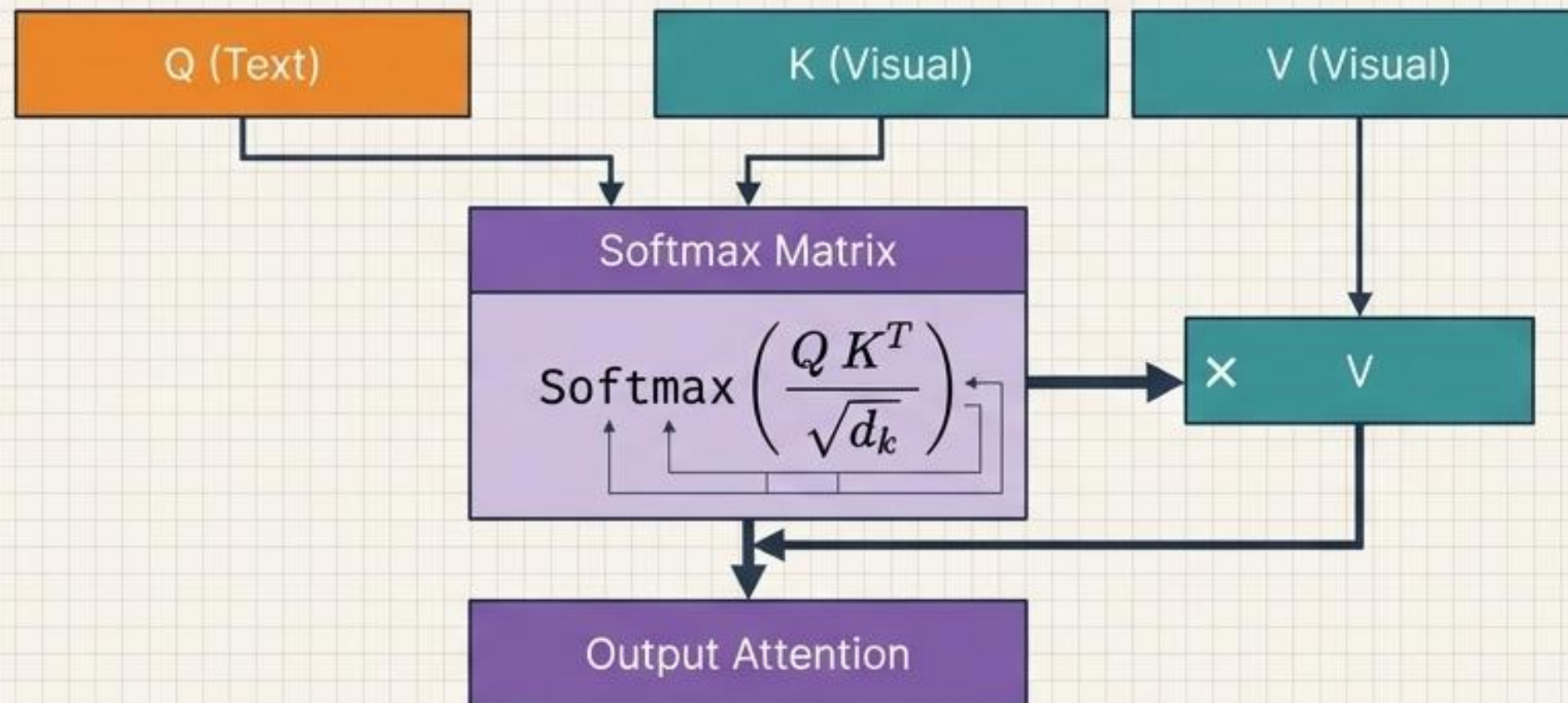
Uniform temporal sampling divides untrimmed videos into exactly $T=32$ segments. Visual and Text modalities are projected offline into a shared 512-D semantic space, rendering online training highly efficient.

Phase 2: Online Training

Cross-Attention block uses Text to contextualize Vision. The MLP Head maps 512-D fused features to anomaly scores between 0 and 1.

Forcing Semantic Evaluation via Text-Queried Visual Attention

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{Q K^T}{\sqrt{d_k}} \right) V$$



Modality Mapping

$Q = T * W_Q$ (Text Query: The semantic concept of the anomaly)

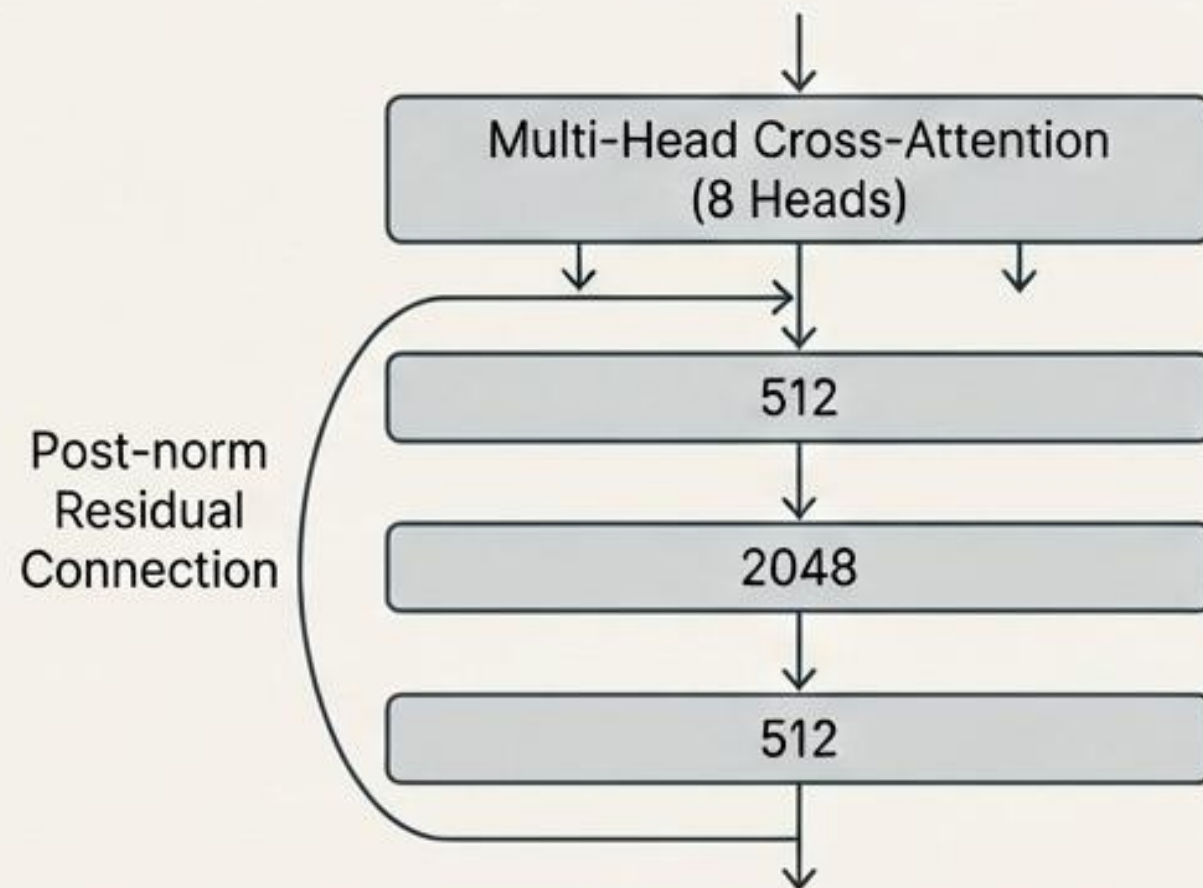
$K = F_{vis} * W_K$ (Visual Key: The raw visual context)

$V = F_{vis} * W_V$ (Visual Value: The features passed forward)

Mechanism Insight

The text query structurally forces the attention weights to prioritize visual features that align with semantic descriptions of anomalies, mathematically neutralizing pure visual context bias.

Micro-Architecture Specifications and MIL Ranking Constraints



Attention Block Topography:

Type: Multi-Head Cross-Attention

Heads: 8 ($d_k = 64$ per head)

Activation: GELU inside FFN

$$\text{Total Loss} = L_{\text{rank}} + \lambda_{\text{smooth}} * L_{\text{smooth}} + \lambda_{\text{sparse}} * L_{\text{sparse}}$$

Top-K Ranking (K=8):

Forces top 8 anomalous segments higher than top 8 normal segments by a margin of 1.0.

Smoothness ($\lambda=8e-5$):

Penalizes erratic score jumps between adjacent frames.

Sparsity ($\lambda=8e-5$):

L1 penalty enforcing the prior that anomalies are rare events.

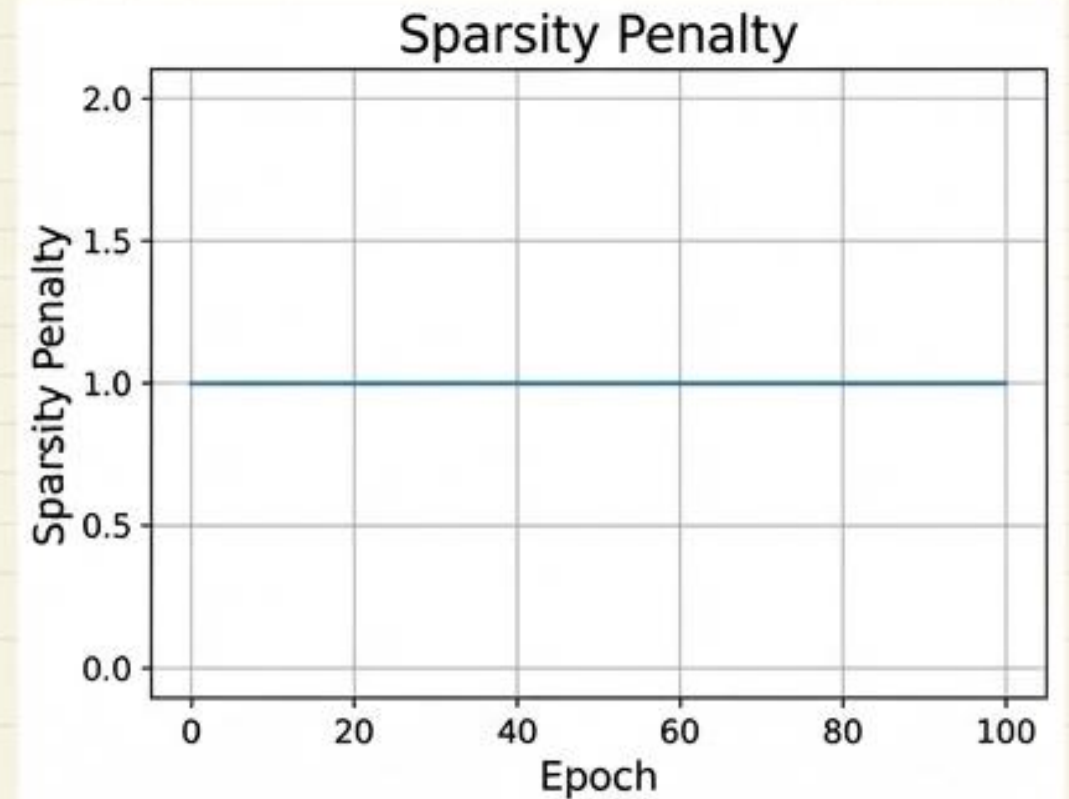
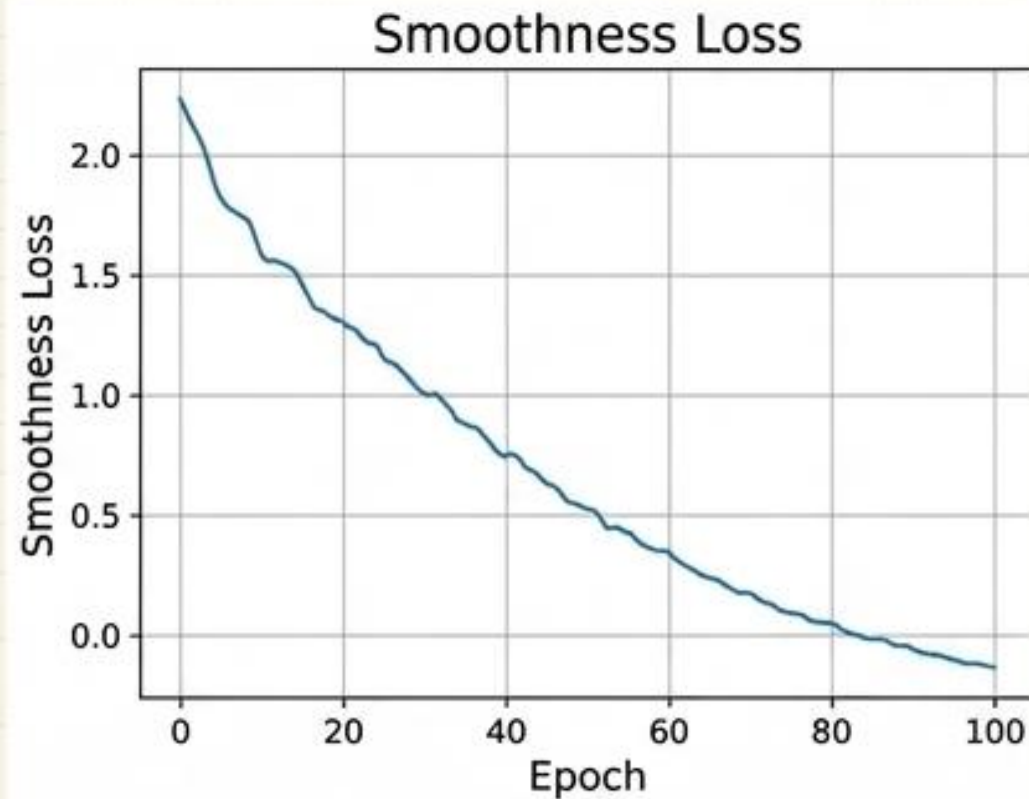
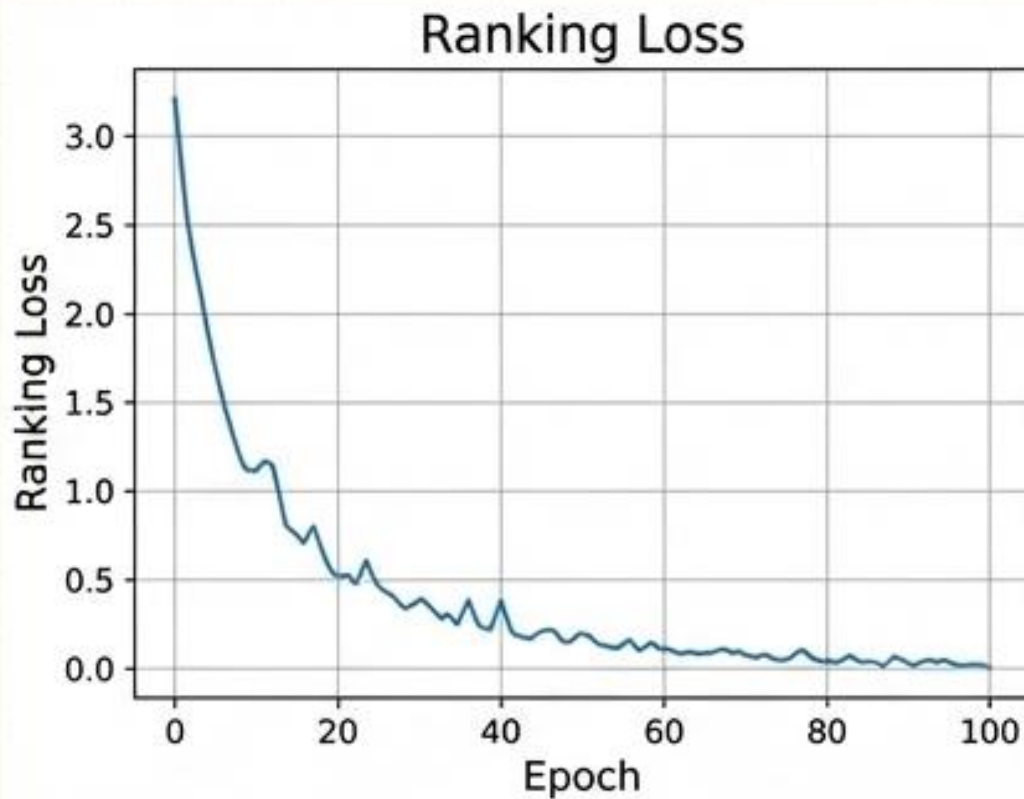
Validating V1 Efficacy Against Foundational and SOTA Models

Method	Supervision	Frame-Level AUROC	Model Size
Sultani et al. (CVPR 2018)	Weak	75.41%	~30M (C3D)
OURS (V1)	Weak	77.14%	2.17M
RTFM (ICCV 2021)	Weak	84.30%	24.72M
MGFN (2022)	Weak	86.98%	28.65M
Sun et al. (TMM 2024)	Weak	88.73%	–

Analytical Takeaway

We achieved a +1.73% absolute improvement over the foundational MIL baseline while utilizing less than 10% of the parameter count, empirically proving the high efficacy of language guidance over brute-force visual scaling.

Optimization Dynamics: Convergence, Coherence, and Sparsity



Optimization Behavior:

- **Ranking Loss:** Shows rapid initial descent and clear convergence on ranking objectives.
- **Smoothness Loss:** Demonstrates a steady decrease, promoting temporal coherence in scores.
- **Sparsity Penalty:** Remains flat, indicating it quickly satisfies the constraint and has minimal impact after early epochs.

Auditing V1: Architectural Blind Spots

Limiting SOTA Performance

Deficiency Matrix

Gap 1: Missing Feature Magnitude

Status in V1: Discards raw visual features post-attention.

Literature Gap: RTFM and MGFN prove the raw L2-norm is a critical, independent indicator of anomaly intensity.

Gap 2: Fixed, Noisy Top-K Selection

Status in V1: Hardcodes K=8 selection from epoch 1.

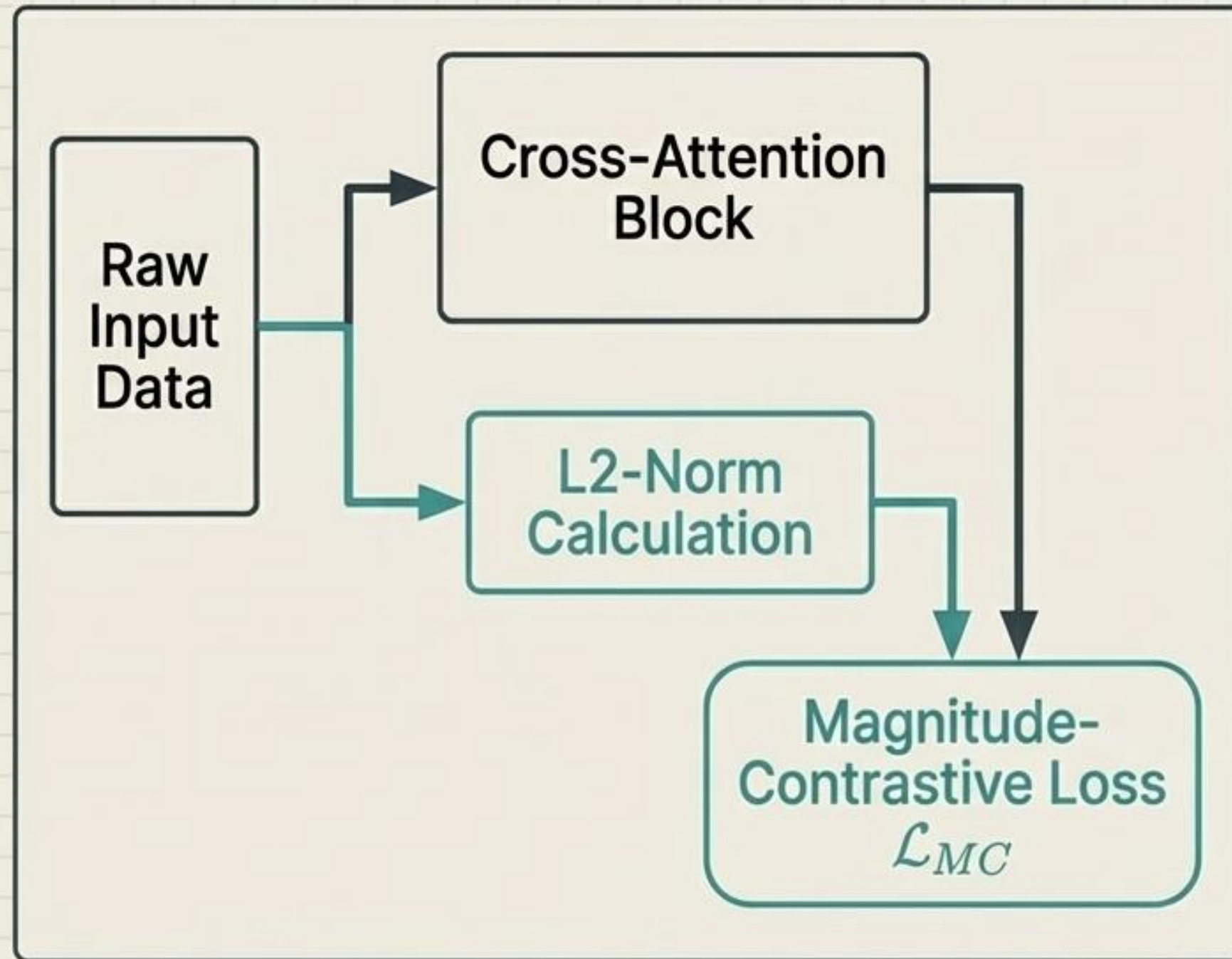
Literature Gap: Early in training, 6 of 8 selected segments are likely noise, which severely destabilizes gradients.

Gap 3: Static Weak Labels

Status in V1: Constrained by video-level binary labels for all 100 epochs.

Literature Gap: SOTA models (MIST) dynamically bootstrap pseudo-labels to achieve quasi-strong supervision.

V2 Roadmap: Integrating a Temporal Feature Magnitude Branch



Priority 1: Feature Magnitude Branch

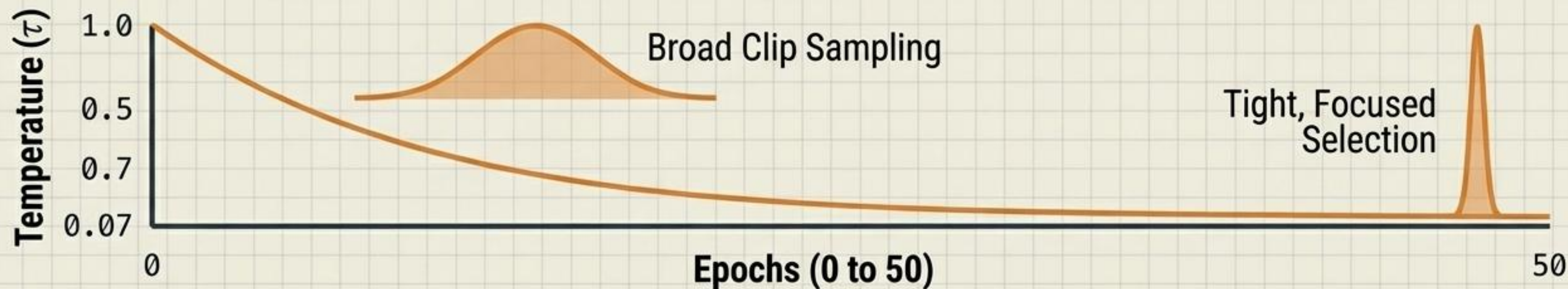
Expected AUROC Gain: +5-7%

Literature Anchor: RTFM (Tian et al., ICCV 2021) & MGFN

Mechanism: Instead of discarding raw visual features after text-guided attention, we compute the L2-norm of the raw visual vectors directly.

The Logic: Anomalous events (e.g., explosions, fleeing crowds) inherently generate higher visual intensity (magnitude) than static backgrounds. We will append a magnitude-contrastive loss term to the total loss landscape.

V2 Roadmap: Replacing Hard Top-K with Adaptive Instance Selection



Priority 2: Adaptive Instance Selection (AIS)

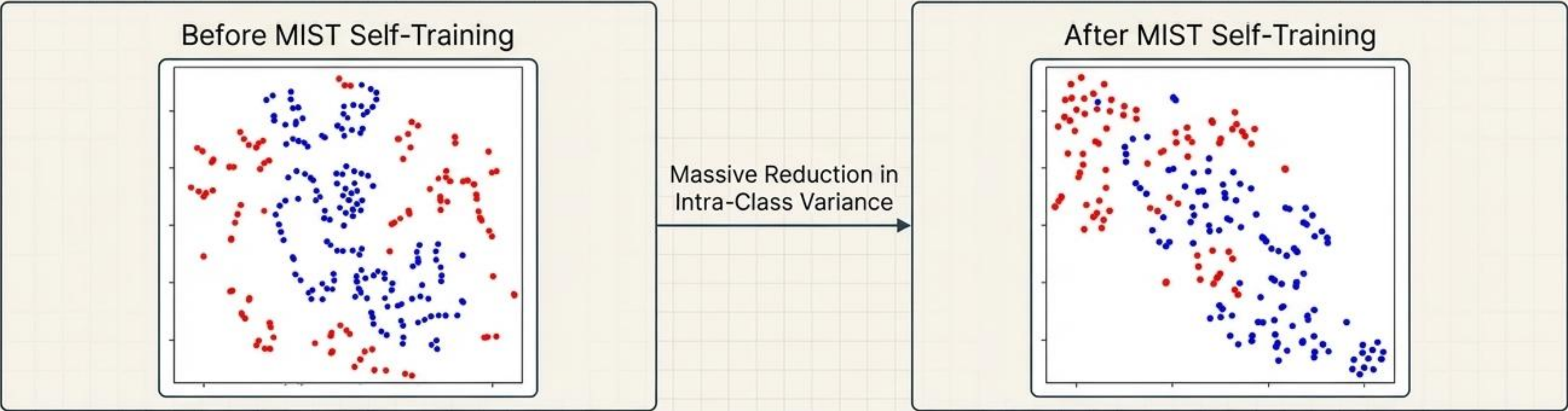
Expected AUROC Gain: +2–4%

Literature Anchor: Lightweight AIS & UMIL

Mechanism: Replace the rigid $K=8$ MIL selection with a soft, uncertainty-weighted Softmax selection over all $T=32$ segments.

The Logic (Curriculum Learning): Implement temperature annealing. Early in training, the model samples broadly across the video. As confidence grows (temperature drops), attention tightly focuses on specific clips.

V2 Roadmap: Bootstrapping Supervision via MIST-Style Self-Training

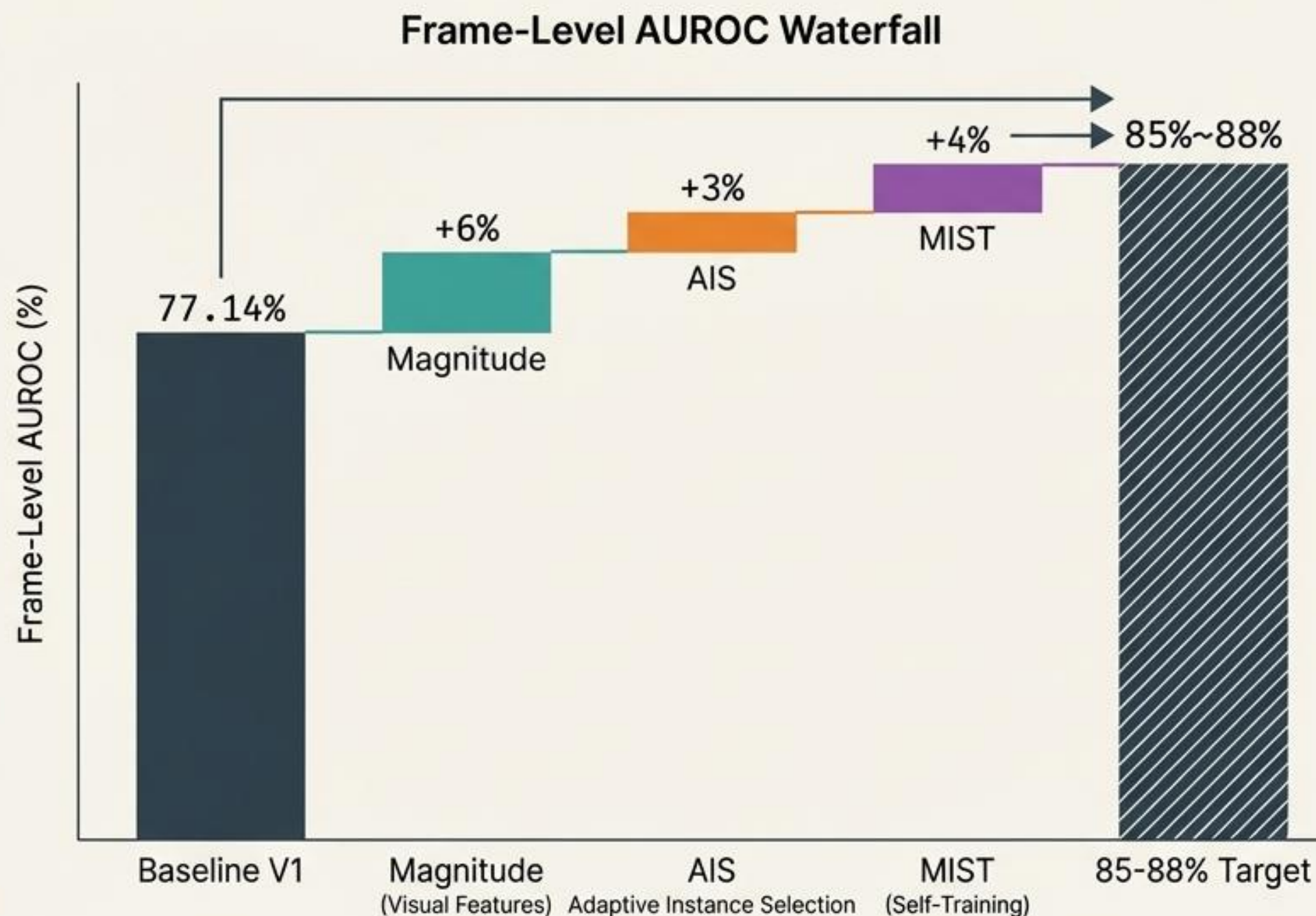


Priority 3: MIST-Style Self-Training (Expected Gain: +3-5%)

Mechanism: Phase 1 trains via standard MIL. Phase 2 uses the model's own Top-K predictions to generate frame-level pseudo-labels, switching to a rigorous **Binary Cross-Entropy (BCE)** loss.

Component	Files to Modify	Technical Difficulty
Feature Magnitude	model.py, loss.py	Low
Adaptive Selection	loss.py, train.py	Medium
Self-Training	train.py, dataset.py	High (Two-stage loop)

The Waterfall to 88% AUROC and Beyond Euclidean Space



The Path to State-of-the-Art

By sequentially implementing these literature-backed upgrades, we project a confident, data-driven trajectory toward the 85-88% target ceiling for UCF-Crime.

Future Horizons (Thesis Exploration)

- **Hyperbolic Space Geometry:** We currently map features in Euclidean space. Future architectures may map to a Poincaré ball (hyperbolic space) where distances grow exponentially, forcing radical separation of visually similar normal/anomaly pairs.
- **Late-Fusion Modalities:** Integration of Optical Flow and Audio streams.

Key References & Source Literature

Foundational Base:

- Sultani, W., et al. (2018). Real-world Anomaly Detection in Surveillance Videos. CVPR.
- Radford, A., et al. (2021). Learning Transferable Visual Models From Natural Language Supervision (CLIP). ICML.

Architectural Upgrades:

- Tian, Y., et al. (2021). Weakly-supervised Video Anomaly Detection with Robust Temporal Feature Magnitude Learning (RTFM). ICCV.
- Li, J., et al. (2022). MIST: Multiple Instance Self-Training Framework. CVPR.
- Lv, H., et al. (2023). Unbiased Multiple Instance Learning (UMIL). CVPR.

State-of-the-Art Language Guidance:

- Sun, H., et al. (2024). Enhancing Weakly Supervised Multimodal Video Anomaly Detection through Text Guidance. IEEE TMM.
- Zanella, R., et al. (2024). Harnessing Large Language Models for Training-free Video Anomaly Detection (LAVAD). CVPR.